



Department of Computer Science & Information Technology



Course Information

Course Code: **CS-08519**

Credit Hours: **4 (3+1)**

Lab Hours/Week: **3**

Course Title: **Computer Networks**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

a. Lab Objectives – Computer Networks

- Enable students to understand core networking concepts and perform basic troubleshooting to ensure network connectivity and reliability.
 - Develop practical skills in configuring, testing, and maintaining computer networks using simulation tools like Cisco Packet Tracer.
 - Develop the ability to design and implement efficient IP addressing schemes and subnetting.
 - Explore **switching concepts** for establishing and managing Local Area Networks (LANs) and VLANs
 - Understand and configure **routing protocols** for data forwarding and inter-network communication.
 - Implement and manage **Access Control Lists (ACLs)** for controlling network access and traffic flow.
 - Understand and apply fundamental **network security mechanisms** for protecting data and infrastructure.
 - Familiarize students with essential network services such as DHCP, DNS, and NAT.
 - Enhance analytical and troubleshooting skills for diagnosing and resolving network performance issues.
 - Prepare students for professional certifications (e.g., CCNA) and industry roles in networking and IT infrastructure.
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